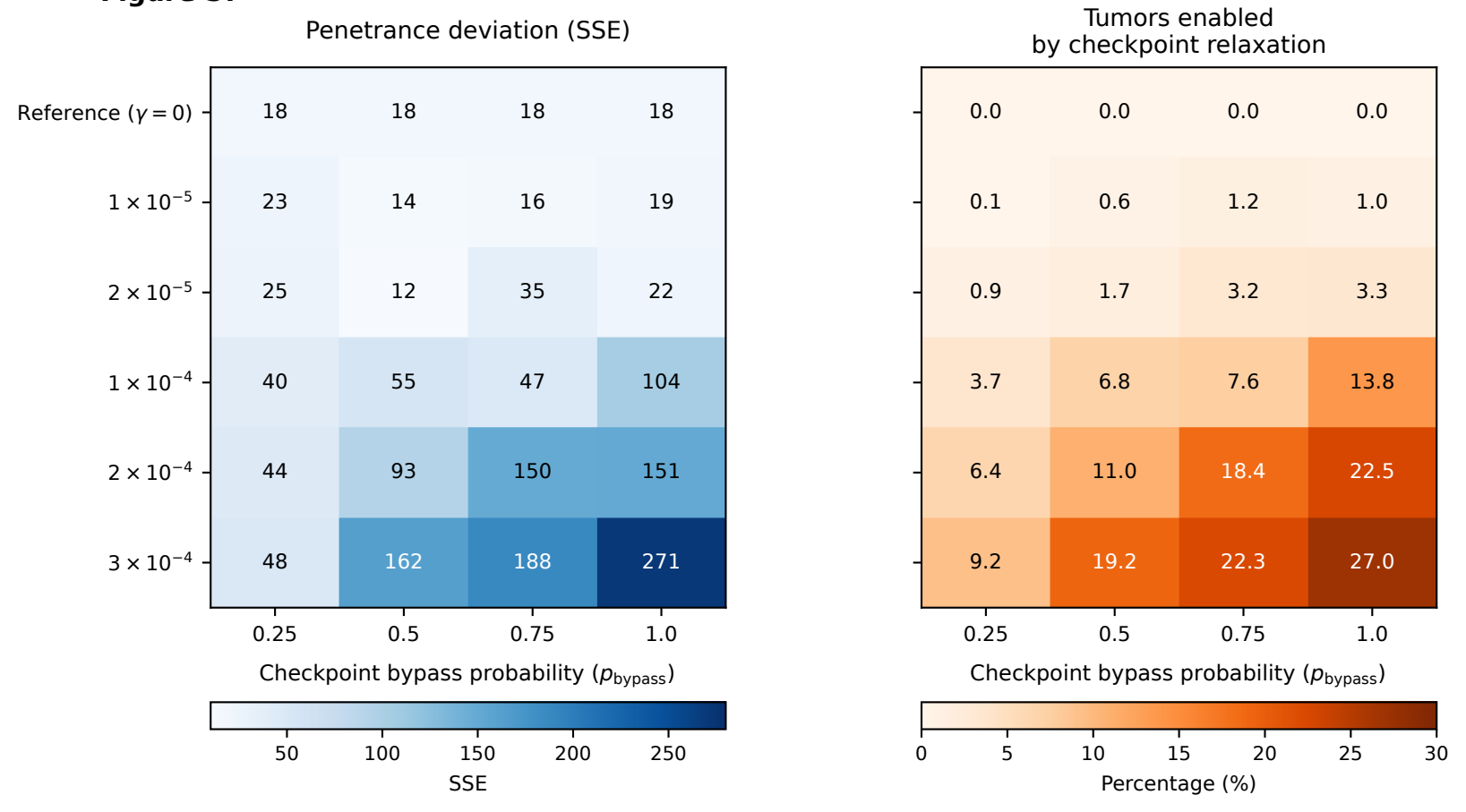
**Figure S7**


Penetrance deviation (SSE)
Reference ($\gamma = 0$)
$1 \times 10^{-5}$
$2 \times 10^{-5}$
$1 \times 10^{-4}$
$2 \times 10^{-4}$
$3 \times 10^{-4}$
18 18 18 18
23 14 16 19
25 12 35 22
40 55 47 104
44 93 150 151
48 162 188 271
0.25 0.5 0.75 1.0
Checkpoint bypass probability ($p_{\text{bypass}}$)
50 100 150 200 250
SSE
Tumors enabled
by checkpoint relaxation
0.0 0.0 0.0 0.0
0.1 0.6 1.2 1.0
0.9 1.7 3.2 3.3
3.7 6.8 7.6 13.8
6.4 11.0 18.4 22.5
9.2 19.2 22.3 27.0
0.25 0.5 0.75 1.0
Checkpoint bypass probability ($p_{\text{bypass}}$)
0 5 10 15 20 25 30
Percentage (%)